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ACCOUNTING FOR PRIMARY PRODUCTION CHANGES IMPROVES THE SKILL OF A BRAZILIAN CONTINENTAL SHELF FISHERY ECOSYSTEM MODEL TO UNDERSTAND CHANGES IN BLACKFIN GOOSEFISH (*Lophius gastrophysus*)

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INTRODUTION

Global environmental changes can affect particular populations and entire biological communities, but the intensity of effects is mediated by species sensitivities and tolerances to direct and indirect effects



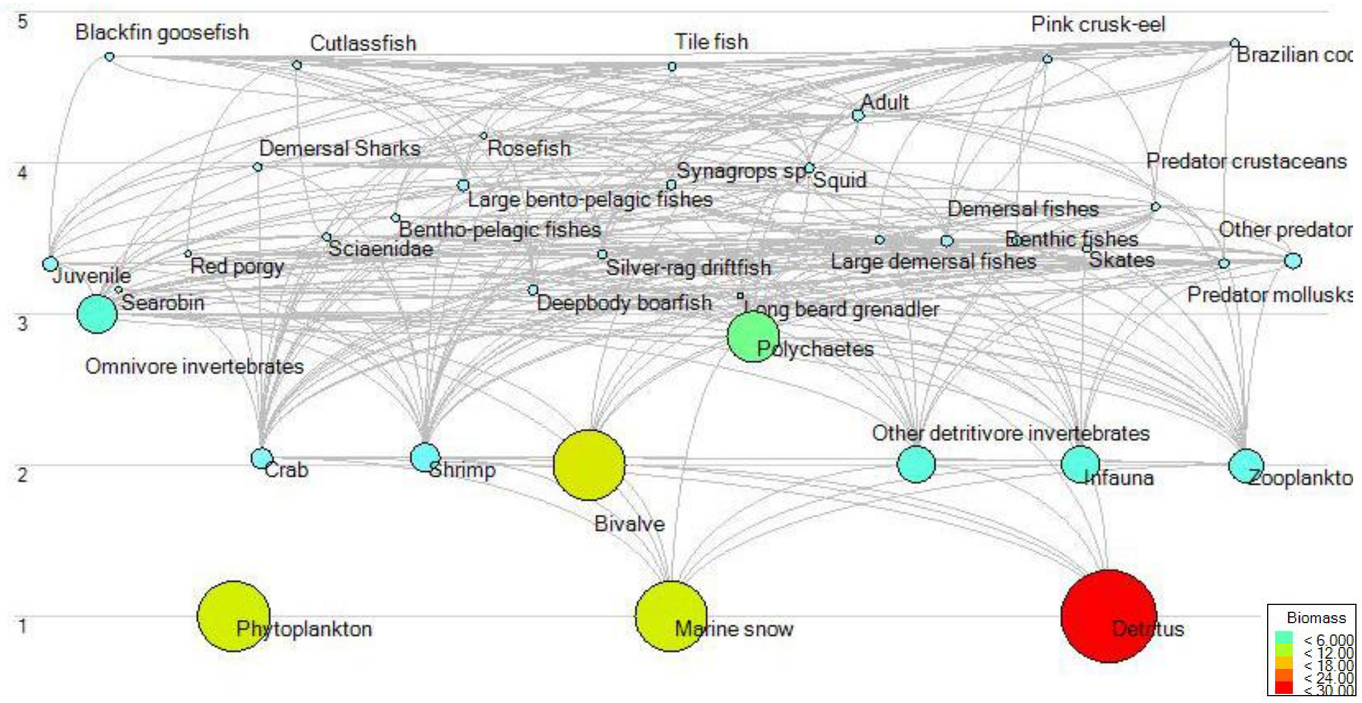


Fig 1. The initial model - Nascimento *et al.* 2012

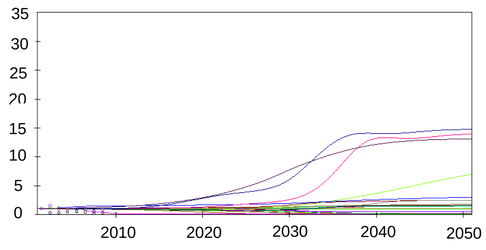


Figure 2. Nascimento *et al.* (2012) model projection to 2050 including the fishing time series, with no temperature or primary production increase simulation

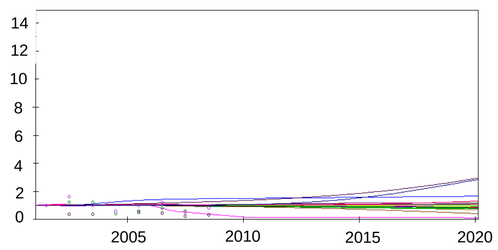


Figure 3. Detail of Nascimento *et al.* (2012) model projection to 2050 including the fishing time series, with no temperature or primary production increase simulation

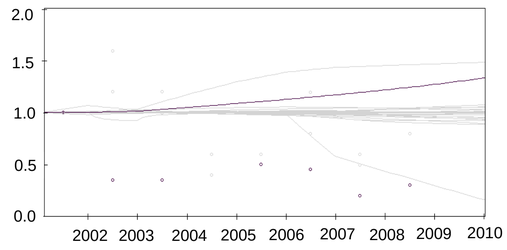


Figure 4. A detail of Nascimento *et al.* (2012) model projection to 2050 including the fishing time series, with no temperature or primary production increase simulation, highligning the time series (dots) and the predicted biomass of Backfin goosefish

MATERIAL AND METHODS

In this pilot exploration, we examined the effects of projected global changes in primary production on a Brazilian continental shelf food web model constructed by Nascimento *et al.* (2012 – Figure 1) given three scenarios of global temperature change. At this stage, these explorations were conducted without knowledge of local and spatial heterogeneity and without explicit knowledge of tolerances and sensitivities of particular functional groups other than how changes in primary production cascades through the present model tropically.

We examined three scenarios of changes in primary production, two of them extreme and one intermediate (Princeton/Marra *et al.* (2003) (1.096), Princeton/Dunne (1.046) and Princeton/Poly (0.989). For each value, we made the simulations using the global changes scenario of lower (1.4°C) high (5.8°C) and intermediate (3.6°C) temperature increases. We ran the simulations for 50 years with and without the global change scenarios, while maintaining the current level of fisheries

RESULTS AND DISCUSSION

All simulations showed the extinction of Blackfin goosefish, a recent target specie in Brazilian waters, among the 50 simulated years. This species was especially sensitive to the specified global changes. The rapid extinction of Blackfin goosefish fits more with the changes in primary production than to the temperature or just the fisheries. For Blackfin goosefish, accounting for global changes in productivity leads to the best fit of the simulated trajectory with the observed trajectory over the last 10 years. This is especially true for the simulation with the largest increase in primary production.

ACKNOWLEDGMENTS

Villy Christensen, Marta Coll and Cecilia Amaral for their technical help, CAPES and CNPq commissions, from Brazil, that supplied the first author with scholarships. TAO thanks the Pew Fellows Program in Marine Conservation, Pew Environment Group

Princeton/Poly

Princeton/Dunne

Princeton/Marra 2003

BEST SCENARIO

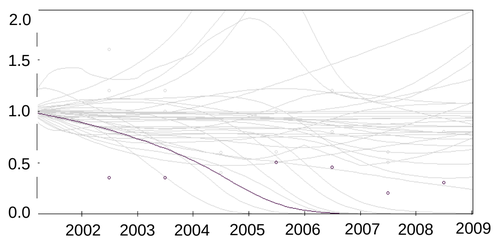


Figure 5. Simulation including the Princeton/Poly scenario of primary production, and a increase of **1.4°C**

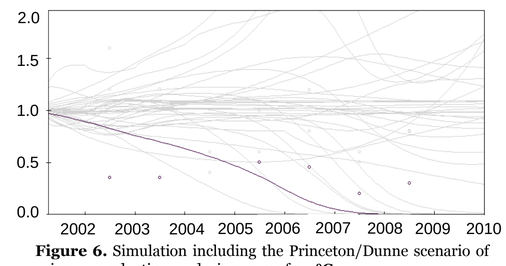


Figure 6. Simulation including the Princeton/Dunne scenario of primary production, and a increase of **1.4°C**

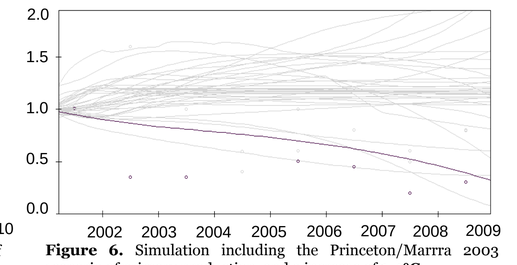


Figure 6. Simulation including the Princeton/Marrra 2003 scenario of primary production, and a increase of **1.4°C**

SCENARIO INTERMEDIATE

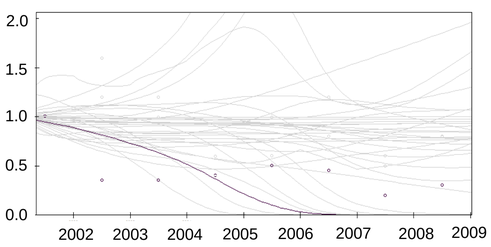


Figure 7. Simulation including the Princeton/Poly scenario of primary production, and a increase of **3.6°C**

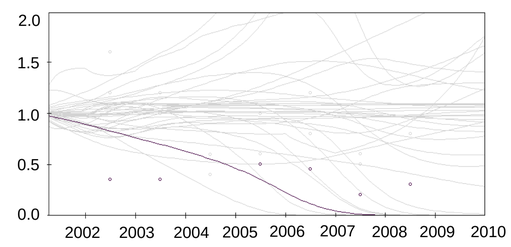


Figure 8. Simulation including the Princeton/Dunne scenario of primary production, and a increase of **3.6°C**

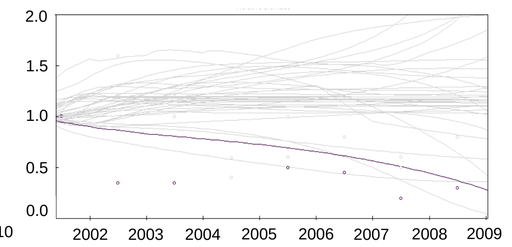


Figure 9. Simulation including the Princeton/Marra 2003 scenario of primary production, and a increase of **3.6°C**

WORST SCENARIO

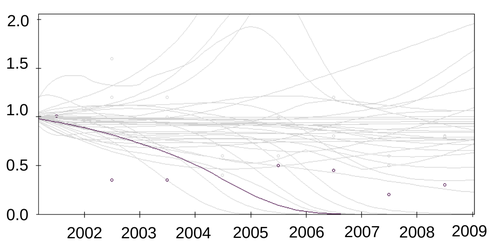


Figure 10. Simulation including the Princeton/Poly scenario of primary production, and a increase of **5.8°C**

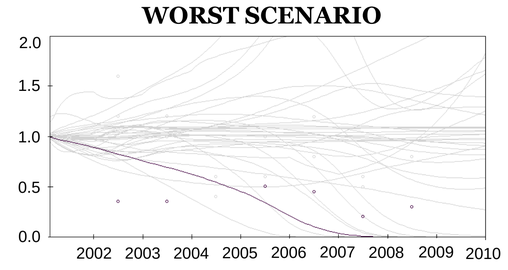


Figure 11. Simulation including the Princeton/Dunne scenario of primary production, and a increase of **5.8°C**

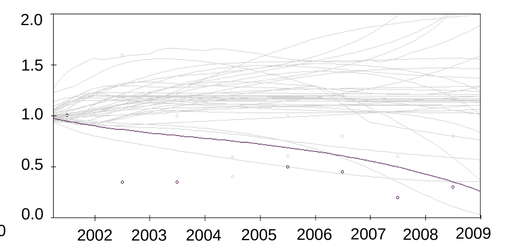


Figure 11. Simulation including the Princeton/Marra 2003 scenario of primary production, and a increase of **5.8°C**